Reference Occupancy Schedules and Synthetic Signal Injection in Building Energy Modelling

Baseline Occupancy and Internal Load Assumptions across Regulatory Codes

Building energy simulation codes utilize standardized baseline inputs to ensure a consistent framework of comparison when validating the performance of high-performance design variants against code-compliant reference buildings.[1] The core parameters governing internal heat gains and operational schedules are derived from major regulatory standards, including ASHRAE Standard 90.1 Appendix G, the United States Department of Energy (DOE) Commercial Prototype Building Models (developed by Pacific Northwest National Laboratory, or PNNL), and Canada's National Energy Code for Buildings (NECB 2017 and 2020).[3] Under these frameworks, occupant behavior and the associated internal loads are modeled deterministically using peak densities—such as occupant density, lighting power density (LPD), and receptacle or plug load density—multiplied by fractional schedules or diversity factors that vary by day type.[6]

Under ASHRAE 90.1 Appendix G and the linked PNNL prototype models, the baseline performance is fixed at a stable level of performance approximately equal to the 2004 standard, allowing buildings of any era to be rated using a unified Performance Cost Index (PCI) and Building Performance Factors (BPFs).[2] Concurrently, Canada's NECB relies on an objective-based Part 8 performance path that compares the annual energy consumption of a proposed design to that of a reference building modeled to prescriptive standards.[1] In the Canadian context, Part 8 specifies standard operating schedules, peak occupant densities, and default receptacle densities in Table 8.4.4.2 (historically Table 8.4.3.2) to maintain modeling consistency across different jurisdictions.[9]

For example, an Office archetype under NECB 2017/2020 dictates a peak occupant density of $25.0\ \mathrm{m^2/person}$ ($0.040\ \mathrm{people/m^2}$), a peak receptacle load of $7.5\ \mathrm{W/m^2}$, and operating Schedule A.[9] A School (K-12) archetype requires a much tighter occupant density of $8.0\ \mathrm{m^2/person}$ ($0.125\ \mathrm{people/m^2}$) and a peak receptacle load of $5.0\ \mathrm{W/m^2}$ under operating Schedule D.[9] Multi-unit residential buildings assume $25.0\ \mathrm{m^2/person}$ ($0.040\ \mathrm{people/m^2}$) and $5.0\ \mathrm{W/m^2}$ under operating Schedule G.[9]

These defaults act as the baseline reference in the absence of better site-specific information, although compliance paths permit alternative schedules if they remain identical between the proposed and baseline simulations.[6] The table below outlines the peak densities and fractional schedule behaviors for office, retail, and hotel uses across these baseline standards.[4]

Building Use Type	Standard / Archetype	Peak Occupant Density	Peak Lighting Power Density (LPD)	Peak Receptacle / Plug Load Density	Weekday Fractional Schedule (Core Occupied Hours)	Weekend Fractional Schedule (Core Occupied Hours)
Office	ASHRAE 90.1 / PNNL Prototypes [7]	0.040 peo ($25.0\ \mathrm{m}^2/$)	$6.5 -$ (varies by vintage) [14]	$8.0 -$ ($0.75 -$) [15]	0.85 - 0.95 (08:00 - 17:00); drops to 0.05 - 0.10 overnight [6]	0.00 - 0.05 (Unoccupied/night baseline) [7]
Office	Canada NECB 2017/2020 [4]	0.040 peo ($25.0\ \mathrm{m}^2/$)	$10.0\ \mathrm{W}/$ (prescriptive limit) [4]	$7.5\ \mathrm{W}/\mathrm{m}$ [9]	Schedule A: 0.85 - 0.95 (09:00 - 17:00); 0.05 overnight [6]	0.00 - 0.05 (Schedule A weekend baseline) [6]
Retail	ASHRAE 90.1 / PNNL Prototypes [14]	0.100 peo ($10.0\ \mathrm{m}^2/$)	$11.8\ \mathrm{W}/$ ($1.1\ \mathrm{W}/\mathrm{ft}^2$ default) [14]	$2.5 -$	0.70 - 0.90 (09:00 - 21:00); 0.05 overnight	0.80 - 0.95 (Saturdays); 0.10 - 0.50 (Sundays)
Retail	Canada NECB 2017/2020 [4]	0.067 peo ($15.0\ \mathrm{m}^2/$)	$12.0 -$ [4]	$5.0\ \mathrm{W}/\mathrm{m}$	0.70 - 0.85 (Extended retail hours);	0.60 - 0.80 (Extended weekend

					0.05 overnight	operating hours)
Hotel	ASHRAE 90.1 / PNNL Prototypes [13]	$0.051\ \mathrm{peo}$ ($19.5\ \mathrm{m}^2/$)	$4.3 -$ (guestroom focus) [13]	$5.4 -$	0.20 - 0.40 (10:00 - 16:00 occupied dip); up to 1.00 at night [13]	0.30 - 0.50 (Midday dip); up to 1.00 at night [13]
Hotel	Canada NECB 2017/2020 [9]	$0.040\ \mathrm{peo}$ ($25.0\ \mathrm{m}^2/$)	$5.0 -$	$5.0\ \mathrm{W/m}$ [9]	Schedule G: 0.30 (Midday dip); up to 0.90 at night	0.40 - 0.50 (Midday dip); up to 0.90 at night

Methodological Framework: Replacing versus Modulating Baseline Schedules

When researchers and practitioners inject measured or synthetically generated occupancy signals into a whole-building energy model, they operate under two distinct mathematical and conceptual paradigms: direct schedule replacement, or schedule modulation.[6]

Replacing a schedule entails a complete, sequence-level overwrite of the baseline annual 8760-hour or 8784-hour fractional profile.[20] Under this approach, the standard regulatory diversity factor vector is removed, and a specific time-series vector derived from smart thermostats, badge-swipe counts, Wi-Fi connections, mobile positioning GPS data, or predictive machine learning models is directly assigned to the load objects.[12]

Modulating a schedule, conversely, preserves the foundational mathematical shape of the baseline fractional schedule and its peak code-compliant occupant density.[6] The standard schedule profile is multiplied time-step by time-step by a presence scaling vector or control multiplier.[6] For example, studies assessing occupant behavior uncertainty utilize presence multipliers—such as 0.75, 1.0, and 1.25—applied to standard schedules to evaluate building system robustness without altering the peak load assumptions embedded in the core code of record.[6]

These two approaches serve fundamentally different applications in building energy engineering:

Direct replacement of baseline schedules is necessary for existing building calibration, post-occupancy performance evaluation, and predictive controls.[15] For instance, a study on a medium office building demonstrated that replacing standard DOE prototype schedules with data-mined occupant appliance power consumption profiles revealed actual occupancy rates that were 36.67% to 50.53% lower than standard baseline assumptions.[25] Simulating the corrected, replaced schedule across multiple climate zones prevented significant heating and cooling overestimations, demonstrating how standard profiles overlook operational realities.[25] Replacement is also crucial when evaluating district-level energy systems or urban building energy models (UBEM) where localized occupancy diversity, hybrid work arrangements, or non-standard university classroom timetables drive massive deviations in grid peak demand.[19] Modulation of baseline schedules is standard practice for design-phase regulatory compliance, performance-based energy code filings, and green building certifications.[2] The Canadian BEM roadmap proposes running three parallel simulations—modulating standard schedules by multipliers of 0.75, 1.0, and 1.25—for both the proposed and reference buildings.[6] This approach braskets behavioral uncertainty and reports the worst-case energy performance, preventing designers from gaming the compliance pathway by assuming unrealistically low baseline loads.[6] Modulation allows the model to capture the energy-saving impacts of occupant-centric technologies—such as demand-controlled ventilation (DCV) and occupancy-based light dimming—without distorting the comparative baseline required by the code of record.[4]

Sizing, Ventilation, and Credits: The Regulatory Pitfalls of Overwriting Baseline Densities

Modifying peak occupancy densities directly within a regulatory or compliance-focused energy model introduces systemic errors that break the comparative modeling framework of standards like ASHRAE 90.1 Appendix G and NECB Part 8.[1]

In regulatory energy modeling, compliance is demonstrated by comparing a proposed design model to an automatically generated baseline model.[1] The baseline and proposed models must be subjected to identical internal load intensities and operational schedules to ensure that performance differences are driven strictly by building system efficiencies, envelope characteristics, and energy recovery measures, rather than arbitrary occupancy differences.[1] If a modeler overwrites the design peak occupant densities in the proposed model, the baseline generation tool may fail to match load assumptions, or the modeler may inadvertently fabricate a compliance credit.[4]

At a physical level, overwriting code-of-record occupant densities distorts mechanical system sizing and outdoor air calculations.[29] EnergyPlus and other dynamic simulation engines size HVAC coil capacities and zone design airflows based on peak heating and cooling loads, which include latent and sensible heat gains from occupants.[29] Arbitrarily lowering peak occupant densities reduces internal heat gains, which can cause the simulation's sizing algorithms to undersize cooling systems and cooling towers, leading to simulated unmet hours and thermal discomfort during actual peak summer conditions.[29]

Furthermore, standard commercial ventilation calculations, such as those governed by ASHRAE Standard 62.1 and Canadian provincial building codes, dictate the minimum volume of outdoor air intake based on the design occupant density of each zone [29]:

$$V_{oz} = R_p \cdot P_z + R_a \cdot A_z$$

Where V_{oz} represents the zone outdoor airflow, R_p is the outdoor airflow rate per person, P_z is the zone design occupant peak count, R_a is the outdoor airflow rate per unit area, and A_z is the zone floor area.[29] Overwriting the design peak occupant density directly decreases the simulated minimum ventilation rate during occupied hours.[26] If the peak density is artificially reduced, the baseline HVAC system's outdoor air heating and cooling loads are underestimated, masking envelope thermal losses and yielding a false compliance credit.[4]

Inter-Schedule Coupling Rules and Cascade Control Sequences

Occupancy acts as the primary driving signal for internal building energy loads. In real buildings and high-fidelity simulations, secondary energy-using systems do not operate independently; they cascade downstream of occupant presence.[4] To capture these interactions accurately, modelers must implement robust inter-schedule coupling rules.

Lighting and Daylighting Interactions

Lighting energy schedules must be dynamically coupled to occupancy to reflect modern automatic shut-off and occupancy-sensing controls.[4] Under ASHRAE 90.1-2022, general lighting in offices must be automatically reduced or shut off within 20 minutes of vacancy.[13] Bathroom lighting in hotel guestrooms must shut off within 30 minutes of vacancy.[13]

To simulate this coupling, the instantaneous lighting schedule fraction L_t at any time-step t can be modeled as a function of the occupancy presence vector O_t:

$$L_t = \max\left(O_t \cdot \eta_{\text{sensor}},\, L_{\min}\right) \cdot D_t$$

Where η_{sensor} is a sensor efficiency factor (often modeling a minor control delay), $L_{\min}$ is the minimum lighting fraction required for continuous safety, egress, or corridor security (typically 0.10 to 0.20 of peak power), and D_t is a daylight-responsive dimming multiplier (ranging from 0.0 to 1.0) calculated dynamically by the engine’s illuminance algorithm.[4]

Plug Loads and Standby Baseloads

Unlike lighting, plug loads (computers, printers, servers, and appliances) never drop to absolute zero when occupants depart.[35] Modern commercial buildings carry a substantial, uninterrupted baseload due to equipment remaining in standby sleep modes, network cards maintaining

connections, and unmanaged process loads.[35] Studies on government and corporate office buildings reveal that unoccupied overnight and weekend electricity consumption can exceed 50% of peak operational demand.[36]
To model this, plug load schedules must maintain a non-zero baseload offset:

$$P_{\text{plug},t} = P_{\text{baseload}} + (1 - P_{\text{baseload}}) \cdot O_t$$

Where P_{baseload} is the baseline fraction of power consumed during fully unoccupied periods (typically 0.15 to 0.30 for standard commercial offices) and O_t is the occupancy fraction.[35] If smart receptacle controls are modeled—such as those required by ASHRAE 90.1-2022, which mandates that 50% of receptacles in offices be controlled by occupancy sensors—the P_{baseload} can be dynamically reduced, but a minimum baseline must remain to prevent underestimating equipment power draws.[13]

HVAC Setbacks and Standby Ventilation

When zone occupancy drops to zero (or falls below a designated threshold), terminal units and air handlers should transition to unoccupied standby control sequences.[13] For Variable Air Volume (VAV) reheat terminal boxes, conventional controls often maintain minimum airflows at 30% to 50% of design maximums to satisfy ventilation requirements.[29] When coupled to an occupancy sensor, however, the terminal box enters an "occupied standby" mode, dropping minimum airflow to zero and temporarily widening the thermostat deadband.[29]

The thermostat cooling setpoint $T_{\text{clg},t}$ and heating setpoint $T_{\text{htg},t}$ should adjust dynamically:

$$T_{\text{clg},t} = \begin{cases} T_{\text{clg, occupied}} & \text{if } O_t > 0 \\ T_{\text{clg, setback}} & \text{if } O_t = 0 \end{cases}$$

$$T_{\text{htg},t} = \begin{cases} T_{\text{htg, occupied}} & \text{if } O_t > 0 \\ T_{\text{htg, setback}} & \text{if } O_t = 0 \end{cases}$$

This coupling prevents the simultaneous heating and cooling common in standard VAV reheat configurations and minimizes mechanical outdoor air conditioning when the space is vacant.[30]

Simulation Resolution Conventions and Temporal Down-Sampling Performance Losses

Whole-building energy simulation programs like EnergyPlus accept schedule inputs through various object classes, primarily Schedule:Compact and Schedule:File.[17] Selecting the correct object is essential for maintaining simulation accuracy, particularly when transitioning from

design-day representations to high-resolution measured datasets.[38]
The Schedule:Compact class is standard for prescriptive and compliance-based modeling.[39] It defines fractional values inline within the Input Data File (IDF) using descriptive text strings, date ranges (Through:), day-type identifiers (For:), and clock-time markers (Until:).[17] This format is highly flexible and structured, but it is restricted to deterministic, block-like schedules (e.g., hourly step-changes).[38] Schedule:Compact is computationally efficient for compliance simulations but is impractical for importing continuous, high-resolution empirical datasets.[38]
The Schedule:File class is designed for injecting empirical or synthetically generated high-resolution schedules.[21] This object references an external, comma-separated text file containing 8,760 or 8,784 lines of data corresponding to the hours of a year.[21] Key fields in Schedule:File include Minutes per Item (which must be an integer evenly divisible into 60, such as 1, 5, 10, 15, 30, or 60) and the Interpolate to Timestep flag.[21] When Interpolate to Timestep is set to "No", EnergyPlus applies the interval value directly at each simulation timestep; when set to "Yes", the engine averages between schedule intervals to match the user-specified zone timestep resolution.[21]
In building energy modeling, empirical occupancy data collected at 5-minute or 10-minute intervals is often down-sampled to 30-minute or hourly resolutions to reduce file size and accelerate simulation runtimes.[41] However, this temporal down-sampling introduces several computational and thermal modeling errors.

Peak Load Dampening and Sizing Inaccuracies

Down-sampling occupancy records acts as a low-pass filter, smoothing out high-frequency transients.[19] Real commercial spaces experience rapid, localized occupancy spikes—such as meeting room assemblies, retail checkout lines, or hotel lobby check-ins—where occupancy density rises quickly.[19] Averaging these peaks over a 30-minute or 60-minute window dampens the maximum simulated internal sensible heat gain.[34]
Because thermal mass and mechanical cooling coils respond to the absolute peak sensible loads, down-sampled inputs can underestimate the required local cooling capacity and overlook short-term localized temperature spikes, yielding inaccurate estimates of thermal comfort and zone temperature drift.[29]

VAV Controller and Airflow Dynamics

Dynamic HVAC control loops, including Variable Air Volume (VAV) dampers and variable-speed fan controllers, respond directly to instantaneous zone sensible heat gains and CO2 levels.[29] If the occupancy signal is down-sampled to 30 or 60 minutes, the fast, transient changes in occupant presence are obscured.[32]
In a high-resolution simulation (e.g., 5-minute timesteps), the sudden exit of occupants triggers an immediate ramp-down of VAV supply airflow to minimum levels, saving significant fan and reheat energy.[29] Under a down-sampled schedule, this transition is smoothed over a much longer period, artificially keeping terminal box dampers open and over-ventilating the zone, which leads to an overestimation of heating and cooling energy use.[29]

Performance Evaluation of Predictive Controls

Advanced building control frameworks, such as Model Predictive Control (MPC) and active Demand Side Management (DSM), rely on the alignment of real-time occupancy counts, solar heat gains, and utility pricing signals.[15] High-resolution schedules capture the precise coincidental alignment of peak solar radiation through windows with maximum occupant sensible heat generation.[44]
Down-sampling occupancy signals separates these coincident loads, resulting in inaccurate predictions of peak electricity demands, distorted baselines for demand response programs, and reduced modeled effectiveness of heat-pump load-shifting or pre-cooling controls.[27]

Decision Rules and Strategic Guidelines

To navigate these structural and computational constraints, energy modelers should apply standardized decision rules when structuring load schedules.[4] The table below defines when to modulate or replace schedules by building use, aligning choices with standard citations and the required regulatory or operational modeling objectives.

Building Use Type	**Modeling Objective**	**Recommended Strategy**	**Inter-Schedule Coupling Protocol**	**Domain Precedent & Citations**
Office	Regulatory Compliance (LEED, NECB Part 8, ASHRAE Appendix G compliance)	**Modulate** standard fractional schedules using diversity factor multipliers (e.g., 0.75, 1.0, 1.25).[6]	Keep peak occupant density tied to code; couple lighting with daylight sensors; maintain standard baseline plug load schedules.[4]	Maintains comparative baselines and complies with submittal verification manuals.[6]
Office	Calibrated Energy Retrofits, Model Predictive Control (MPC)	**Replace** baseline schedules with empirical multi-sensor count vectors via	Dynamically override setpoints; drop terminal box airflow rates to zero in unoccupied	Corrects the 20-30% performance gap and maps actual occupant-centric load

		Schedule:File.[15]	zones; dynamically reset VAV static pressures.[29]	waves.[14]
Retail	Compliance and LPD Trade-offs	**Modulate** standard retail occupancy schedules.[16]	Adjust lighting schedules using prescriptive sensor reduction credits; maintain standard peak design occupancy.[4]	Prevents distortions in baseline fan sizing and outdoor air ventilation calculations.[26]
Retail	Grid-Interactive Demand Response, Local Storage Integration	**Replace** with synthetic, high-resolution traffic profiles (10-minute resolution).[27]	Couple store lighting and refrigeration auxiliary power with occupant flow rates; execute precooling algorithms.[27]	Captures thermal mass latency and peak load shedding capacity on the electrical distribution system.[22]
Hotel	Compliance & Code Performance Rating	**Modulate** guestroom occupancy profiles using standard hotel schedule blocks.[13]	Apply prescriptive hotel key-card or occupancy-sensor lighting shutoff schedules; maintain baseline hot water use.[13]	Ensures standardized comparison under the Performance Rating Method.[2]
Hotel	Plant Optimization	**Replace** guestroom	Force guestroom	Resolves the design-vs-actu

	(e.g., Air-Source Heat Pump retrofits)	schedules with historical booking logs and key-card state changes.[13]	fan-coils into unoccupied standby mode during unbooked hours; lower space DHW recirculating loads.[13]	al gap of heat pumps operating under volatile tourist loads.[13]

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